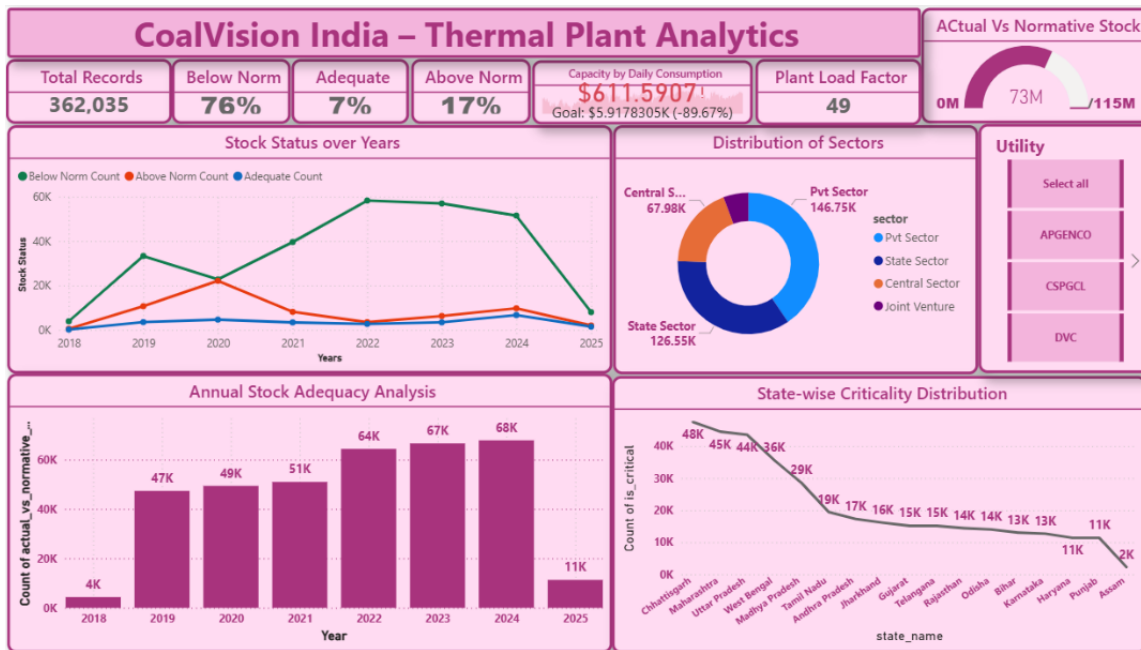


Real-Time Coal Stock Analytics for India's Thermal Power Fleet

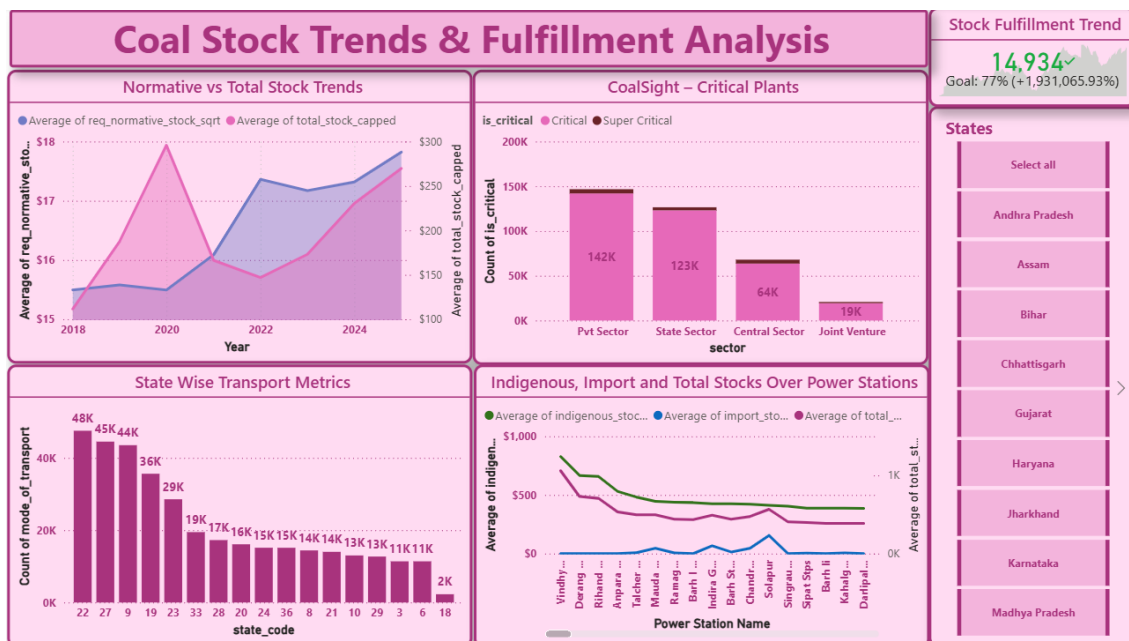
Dashboard Integration

- All charts have been aligned into a single Power BI dashboard by connecting Python scripts for advanced visuals.
- The dashboard combines KPI cards (Capacity, Requirement, Consumption) with stock adequacy charts, transport distribution, and sector/state analysis.
- This hybrid approach leverages Python for custom analytics and Power BI for interactivity.

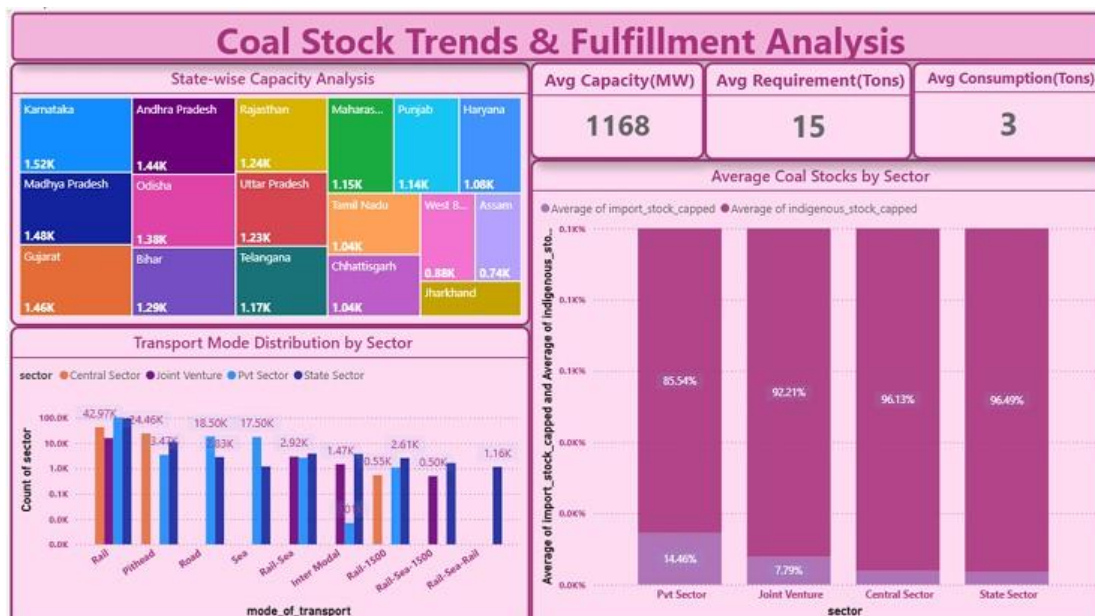
1.Capacity–Requirement–Consumption Dashboard



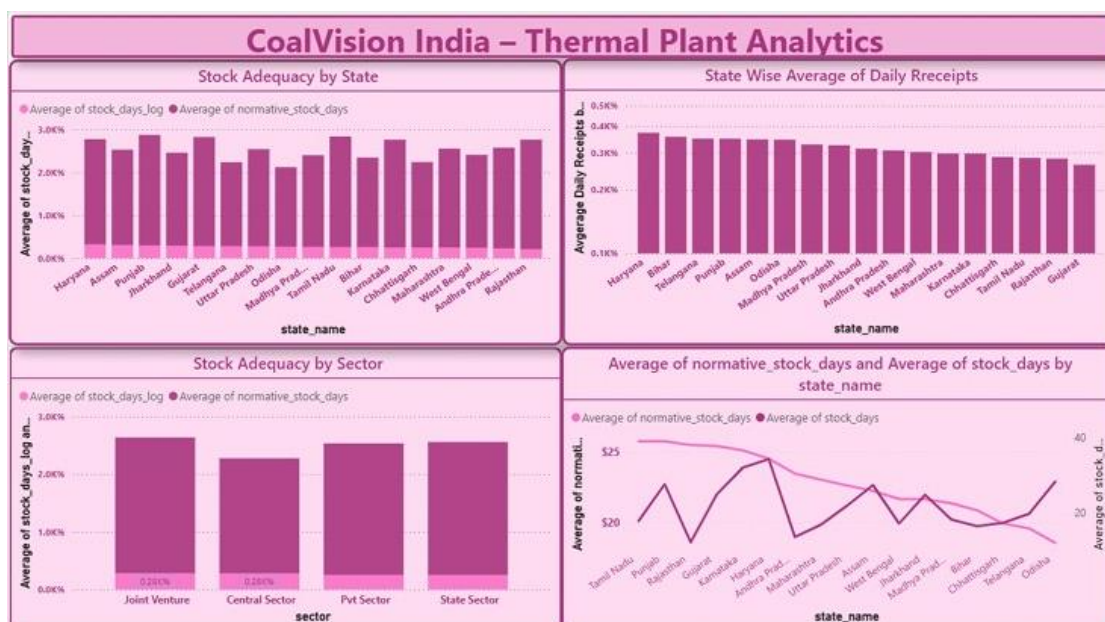
2. Thermal Plant Coal Adequacy & Logistics Dashboard



3. Coal Stock Capacity, Sectoral Mix & Transport Analysis dashboard



4. Coal Stock Adequacy & Daily Receipts Analysis Dashboard



Summary

The patterns show that:

- **Capacity drives requirement and consumption**, but actual consumption often falls short of normative needs.
- **Stock adequacy remains below norm for most plants**, highlighting systemic supply gaps.

- **Rail transport dominates coal logistics**, creating bottlenecks and regional stress zones.
 - **Certain states (Chhattisgarh, Maharashtra, Uttar Pradesh)** consistently face critical stock shortages.
-

Dashboard Conclusion

The dashboard tells a clear story:

- **Top KPI cards** show the dependency chain: Capacity → Requirement → Consumption.
- **Middle visuals** highlight stock adequacy gaps across sectors and states.
- **Transport charts** reveal rail dominance and bottlenecks.
- **Trend lines** show systemic shortfalls but slight improvements in recent years.

Together, the dashboard and insights explain that **India's thermal power ecosystem is expanding capacity faster than coal supply stability**, requiring **strategic interventions in logistics, sourcing, and monitoring** to secure energy reliability.

Final Insights – CoalVision India Dashboard Summary

1. Capacity & Consumption Relationship

- **Average capacity (~1168 MW)** drives both **daily requirement (~15 tons)** and **daily consumption (~3 tons)**.
 - Higher capacity → higher coal demand, but actual consumption remains below normative levels.
 - Indicates **under-utilization or supply constraints**.
-

2. Stock Adequacy Trends

- Across years (2018–2025), **below-norm stock levels dominate (~76%)**.
 - **Adequate or above-norm plants** remain limited.
 - Persistent **shortfalls in coal supply** relative to normative benchmarks, especially in **Private and Central sectors**.
-

3. Transport Mode Distribution

- **Rail** is the primary transport mode (>60%), followed by **Road** and **Pithead**.
 - **Central and Joint Venture sectors** rely heavily on rail logistics.
 - Suggests **dependence on long-haul coal movement** and potential bottlenecks.
-

4. Sector & State Insights

- **State Sector** shows the most balanced stock adequacy.
 - **Private Sector** faces higher criticality.
 - **Chhattisgarh, Maharashtra, and Uttar Pradesh** lead in critical stock counts → **regional stress zones**.
-

5. Normative vs Total Stock

- **Normative stock requirements consistently exceed actual totals.**
 - Reveals a **systemic gap in stock fulfillment**.
 - Gap narrows slightly post-2023 → **incremental improvement in supply efficiency**.
-

Data-Driven Discoveries

- **Stock adequacy:** Majority of plants operate **below normative stock levels (~76%)**, with only a small share adequate or above norm.
 - **Sectoral stress:** Private and Central sectors face higher criticality, while State sector shows relatively balanced adequacy.
 - **Regional hotspots:** States like **Chhattisgarh, Maharashtra, and Uttar Pradesh** consistently report critical stock conditions.
 - **Transport dependency:** Heavy reliance on **rail logistics** creates bottlenecks, limiting flexibility in coal movement.
 - **Normative vs actual stock:** Systemic gap persists, though marginal improvements are visible post-2023.
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Strategic Implication

To ensure reliable power generation, focus must shift toward:

- **Optimizing rail logistics** and diversifying transport modes.
 - **Strengthening indigenous coal sourcing** to reduce import reliance.
 - **Implementing dynamic stock monitoring** for real-time adequacy tracking.
 - **Exploring predictive modelling** to forecast demand-supply mismatches.
 - **Automate analysis pipelines** for faster updates and stakeholder reporting.
-

Overall Conclusion

India's thermal power ecosystem shows **capacity growth outpacing coal supply stability**.

While logistics and stock management have improved marginally, **critical stock conditions persist**, especially in high-capacity states. Strategic focus should shift toward:

- **Rail-logistics optimization**
 - **Indigenous sourcing**
 - **Dynamic stock monitoring** to achieve **normative adequacy across all sectors**.
-

 Overall, the analysis underscores the need for **proactive coal stock management and supply chain resilience** to align actual operations with normative requirements and secure India's energy future.